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Building engineering safety risk assessment and early warning mechanism construction based on distributed machine learning algorithm

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ABSTRACT

Cloud computing has become a hot topic in the industry and academic circles. Large-scale data processing is realized by centralizing system management of network resources, computing resources, and storage resources. Among them, the distributed computing platform greatly improves the productivity of programmers by abstracting the implementation details of distributed computing, and is widely used in building safety assessment. In this study, the construction safety evaluation index system was established for the construction site of common accidents, and the index weight was determined by the analytic hierarchy process. For the safety index evaluation task, an early warning mechanism based on distributed computing and extension theory to build cloud security management is established, which saves the overall calculation time and quantitatively evaluates the security status of the construction site. Finally, the method is verified by engineering examples, and the results show that the method is practical and effective. The method is convenient for computer programming, easy to operate and implement, and has strong applicability. Based on distributed computing and extended cloud model, not only can the overall state of construction safety be judged, but also potential security problems can be identified based on information feedback. The objective existence of hazards in the construction process can be grasped, which has important guiding significance for the construction process.

1. Introduction

Cloud computing is a combination of traditional network technology and computer technology. The cloud computing platform hides complex hardware and software technologies from users and provides services such as storage, computing, and software applications (Botta et al., 2016; Chen et al., 2016). Cloud computing technology has been developed for many years, and the technology is maturing. Moving computing and storage from PCs to large data centres has become a trend in IT technology (Gai et al., 2016). The distributed system is a system based on interconnected server clusters, which realizes resource sharing and calculation through process communication and task coordination. The combination of parallel computing and distributed computing can realize distributed storage and parallel computing of data (Ali et al., 2015). At present, research in the field of distributed storage and computing has made great progress (Li et al., 2018). The related technologies about cloud computing and distributed computing are very mature, but there is no relevant application in building security early warning.

According to the "2017 National Construction Municipal Engineering Safety Production Accident Report", there were 692 construction municipal engineering safety production accidents in 2017, resulting in 807 deaths, 58 more accidents than in 2016, and 72 casualties. The number Increased 9.15% and 9.80% respectively. In order to better avoid losses, a method must be found to quantify safety management, safety management and assessment results, and to provide early warning when problems arise or are likely to occur, that is, the most popular research safety assessment and early warning mechanism in the field of safety management.

At present, there are some construction safety control measure methods, like grey correlation analysis method, matter-element analysis method and artificial neural network, several methods application integration of construction safety control measure. Although good results have been achieved, the construction safety evaluation is a complex system with many influential factors and dynamic changes, many of which are difficult to quantify (Bond et al., 2017). In addition, the basic data of security inspection is not easy to obtain, resulting in the fuzziness, discreteness and randomness of security evaluation. And extension cloud model combines the characteristics of cloud model uncertainty and the advantages of both qualitative and quantitative

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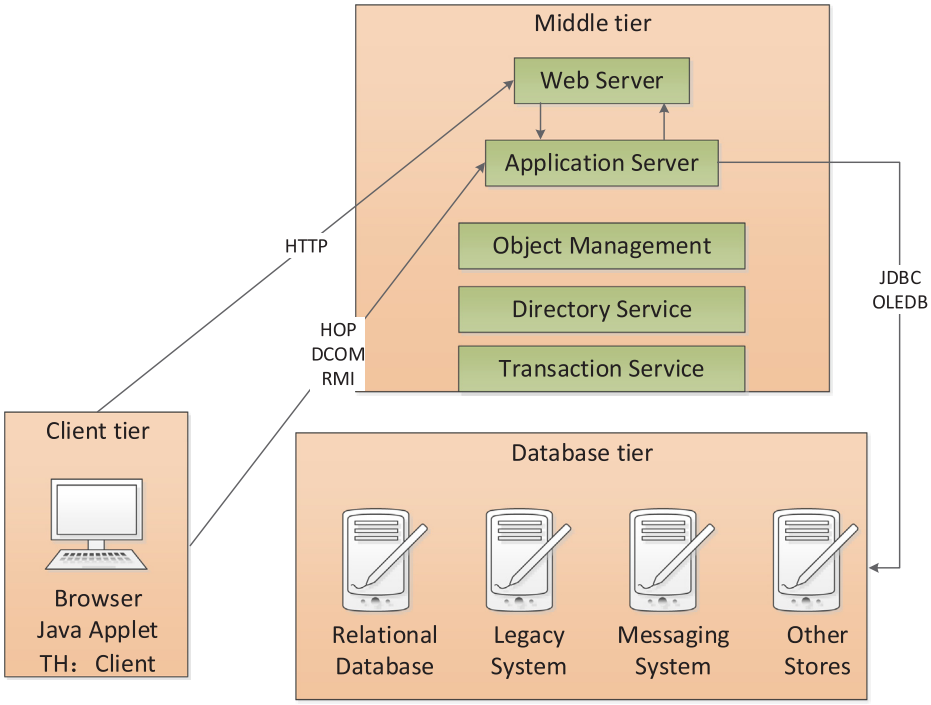

Fig. 1. Tertiary architecture.

analysis of the matter-element theory. Because of this, construction safety can be implemented based on the linguistic value of the mapping between qualitative concepts and numerical said.

Computer Supported Collaborative Work (CSCW) examines how people use computer technology to work together. Distributed computing models are based on distributed object technology, providing a high degree of scalability, reliability, manageability, and flexibility (Lee et al., 2013). When researching construction safety control standards, a large amount of data is required for safety assessment and early warning. Therefore, distributed computing is used to provide users with the necessary collaboration tools to form a running environment for collaborative applications; a service object that provides common collaboration capabilities to develop new applications and integrate existing applications, providing a collaborative application environment that reduces duplication of effort and increases efficiency. Combining the extension cloud model with distributed computing can save the overall calculation time and quantitatively evaluate the safety status of the construction site by using distributed computing. Finally, the method is verified by engineering examples. The results show that the method is practical and effective.

The remains of this paper are organized into four sections. Section 2 contains how the extension cloud theory is adapt to the design of assessment system of security risk control. In Section 3, we build a safety management early warning model for construction enterprises, in order to show the challenges and success when adapting distributed computation into warning model. In the last section we draw conclusion over our research.

2. Assessment system of security risk control based on the extension cloud theory and distributed computation

2.1. Overview of extension cloud theory

As an uncertain model of qualitative and quantitative transformation cloud is, it can be used to describe the construction safety grade limit value. In the theory of matter-element analysis, ordered triples are unified to form the basic elements of things, which are denoted as M = (name of things A, feature D, value E corresponding to feature of things) (Vasile et al., 2015; Göös and Suomela, 2016). In the traditional matter-element model, E represents the threshold value of each index of construction safety evaluation, which is a definite value. However, the actual construction safety grade limit value E has the fuzziness and randomness (Oleinikova et al., 2016).

Cloud is an uncertain model of qualitative and quantitative transformation. Both social science and natural science have proved the universality of normal distribution. The digital features of normal clouds are represented by Fm, Fa and Rg (Wang and Ranjan, 2015). Expected value Fm represents the distribution center of cloud, and it is the point value that best represents the concept of classification level of construction safety. Uncertain degree of the concept of entropy Fa said indicators, which identify the meaning of the construction safety hierarchy between fuzziness, the greater the entropy values shows that the index level of randomness and fuzziness of boundaries, the greater the vice is smaller; super entropy Rg is used to measure the uncertainty of entropy, which is determined by the randomness and fuzziness of entropy. This value reflects the randomness of construction safety index (Arslan et al., 2015; El-Sayed et al., 2018).

Using the normal cloud model, instead of determining the value V, the ordered triples are denoted as "the name of things, the characteristics of things, and the cloud value". The extension cloud model is denoted as:

$$M = \begin{pmatrix} A_{11} & D_1 & F_{m1}, F_{a1}, R_{g1} \\ \vdots & \vdots & \vdots \\ A_n & D_n & F_{mn}, F_{an}, R_{gn} \end{pmatrix} \tag{1}$$

As a computer model that best reflects the spirit of the Internet, cloud computing will show strong vitality (Singh and Chana, 2016). The cloud can respond quickly to the changing needs of users, that is, ready to use, and is gradually changing the way we work and live from diversity. Cloud computing has greatly saved our manpower and resources and improved the efficiency of our work. Therefore, the introduction of cloud computing in the assessment of building safety risk control can greatly reduce costs and has a high application prospect. At the same time, the application of cloud computing in risk assessment also proposes some improvement requirements for existing cloud

computing models, which will be further elaborated in the subsequent model establishment.

2.2. Overview of distributed computation

Distributed computing is another hot point in computing field. In a distributed computing environment, the structure of an application has changed from a traditional client/server two-tier structure to a three-tier structure of a distributed application, as shown in Fig. 1:

The three-tier structure improves performance and scalability, improves reliability, improves manageability, and increases flexibility and development efficiency. The distributed computing model will solve many drawbacks of the client/server model, and the collaboration has natural distribution. Therefore, the three-level structure of distributed computing naturally becomes the ideal architecture for CSCW applications.

The services in the collaborative work environment are not only the collaboration tools used by the end users, but also are the services of other service objects and applications. Based on this, the building engineering security risk management and evaluation and early warning mechanism based on the extension cloud theory can be developed, in which the theory are adapted to this specific task. It can greatly shorten the computing time and improve the flexible cooperation ability of risk management and evaluation and early warning mechanism.

Fig. 2 depicts the life cycle of a user program on a distributed computing platform. After the user program is submitted, the task is parsed into a set of jobs by the distributed computing platform, such as job #1 to job #n in Figure a. For each job, the distributed computing platform calculates the execution order of the jobs and the dependencies of the data, and explicitly or implicitly describes those using directed acyclic graphs. The decomposition of these user programs is mainly done on the primary node. In particular, the distributed computing platform will distinguish between tasks that have data dependencies on each other (for example, task A needs to use the output of task B as input), and there will be no dependencies between each other, and tasks that can be executed in parallel Identify it. According to the calculation logic, these tasks will be aggregated into a calculation stage, and the related data flow is shown in Fig. 2b. In Fig. 2, there are circles in the job #n, and the directed edges between the circles represent data dependencies.

2.3. Background of architecture early warning

Nowadays, it is particularly important to conduct a special study on the evaluation of construction safety control. Based on the current construction safety management standards, norms and the actual construction site safety management, we build the construction safety control measure model based on extension of cloud, the validity of the model are verified through the engineering case, and the usability. The construction of extension cloud model is shown in Fig. 3.

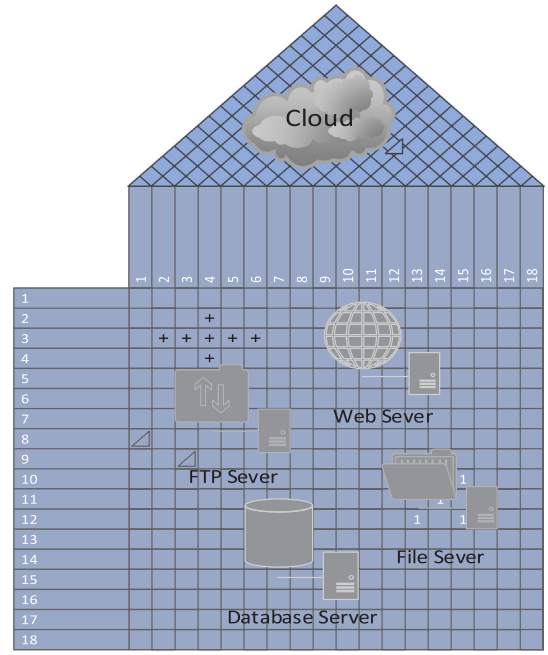

Fig. 3. The construction of extension cloud model.

2.4. Establishment of safety risk control evaluation system

Construction safety risk assessment is a pointer to the construction personnel, management, environment, machinery, equipment safety status of comprehensive evaluation, the rationality of the selected indicators and representation is very important. Enacted according to the country's building construction safety inspection standards for the construction enterprise safety production management regulations, such as analysis of high incidence of accident namely, falling to the construction site, the construction to collapse, objects, mechanical injury, electric shock, the main reason. It is concluded that the main factors influencing the construction safety comes from people, objects, and environment. The construction safety index system is shown in Table 1.

Among them, the score was obtained by the on-site construction

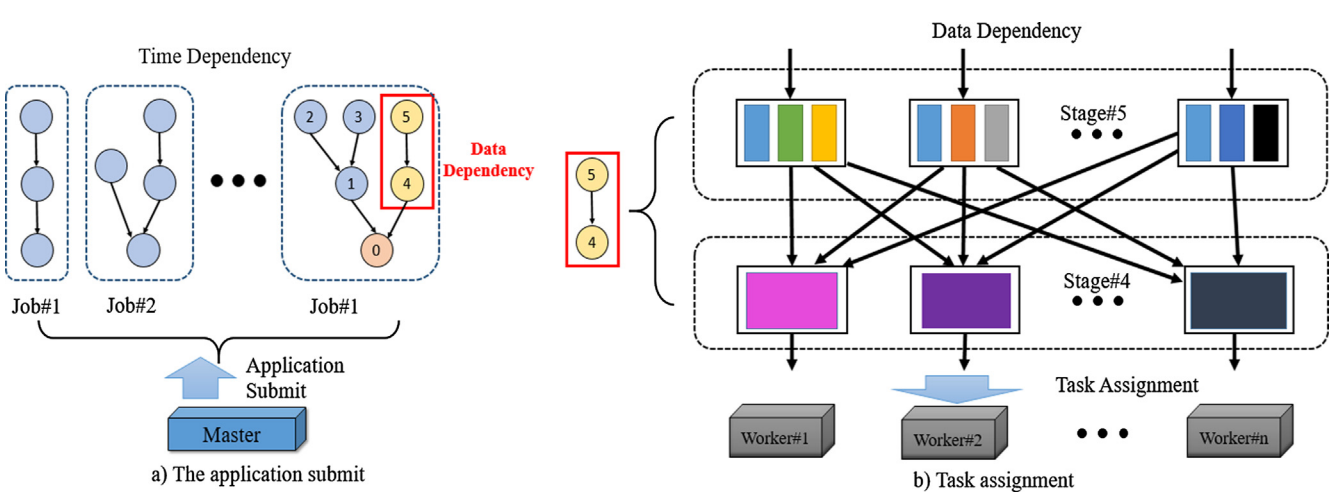

Fig. 2. Data-parallel computing framework: application logic and data movement.

Table 1
Construction safety comprehensive evaluation index system.

First indicators	Second indicators	Third indicators	Evaluation standard
Construction safety evaluation	Human factor RL	Worker safety ideological quality and team spirit knife RL1	Score
		The workers obey the rules and regulations and fulfil their duties RL2	Score
		Active participation rate of workers in safety activities RL3	Ratio
		The psychological quality of workers' safety is RL4	Score
		The literacy of workers RL5	Score
	Factor of substance BL	Safety culture quality and safety technology level of project management BL1	Score
		Safety condition of construction machinery and equipment TF construction materials qualified rate knife BL2	Ratio
		Allocation rate knives for safety protection supplies BL3	Ratio
		Spare rate knives for safety inspection and testing tools BL4	Ratio
		Safety warning mark and slogan setting ratio TF BL5	Ratio
	Environmental factor FL	Engineering technical environment (including geology, hydrology, meteorology, etc.) FL1	Score
		Engineering operation environment (operation area size protection equipment, ventilation lighting, communication, etc.) FL2	Score
		The surrounding environment of the project (the underground pipeline near the project, building structures, etc.)EF FL3	Score
	Management factor SL	The security civilization fund is invested SL1	Score
		Safe education training rate SL2	Ratio
		Safety inspection ratio Yin SL3	Ratio
		Construction organization design SL4	Score
		Special construction plans SL5	Score
		Security technology disclosure SL6	Score
		Hazard identification and control SL7	Score

personnel and relevant experts according to the construction site situation and referring to the safety inspection evaluation form, with a full score of 100 points for each index (Assunção et al., 2015). The ratio is obtained by the ratio of each indicator to the ratio set by the relevant national or local authorities. RL3 is determined by the ratio of the number of times that workers actively participate in safety activities to the total number of times specified by relevant national or local authorities. BL2 is determined by the qualified rate of building materials, and bl3-bl5 is determined by the ratio of the qualified number of construction sites and the total number specified by the state or relevant departments. SL2 shall be determined by the ratio of the number of training required for construction project safety to the number specified by the relevant national or local authorities; SL3 is determined by the ratio of the number of on-site inspections to the total number of inspections specified by the relevant national or local authorities.

2.5. Construction of security risk control evaluation model based on the extension cloud theory and distributed computation

First of all, could be divided into very safe construction safety rating (85–100), security (75–85 points), mild risk (65–75), dangerous (55–65), dangerous (0–55 points), the corresponding level for one to five.

Second, the boundaries of the construction safety classification as a double constraints space processing, expectations of cloud model, entropy and hyper entropy by boundary numerical transformation with the normal cloud model, as shown in formula (2)–(4):

$$F_m = (D_{min} + D_{max})/2 \tag{2}$$

$$F_a = (D_{max} - D_{min})/6 \tag{3}$$

$$R_{\mathrm{gn}} = \mathrm{i} \tag{4}$$

where: R is constant I, which can be adjusted according to the fuzziness, discreteness and randomness of corresponding indicators of construction safety.

Finally, using Eqs. (2)–(4), the standard cloud model of security hierarchy boundary is transformed, as shown in Fig. 4:

Due to the introduction of cloud, the calculation method of correlation degree in general matter-element theory is no longer applicable. The correlation between the content element of a definite value and the content element represented by the cloud model is expressed by the

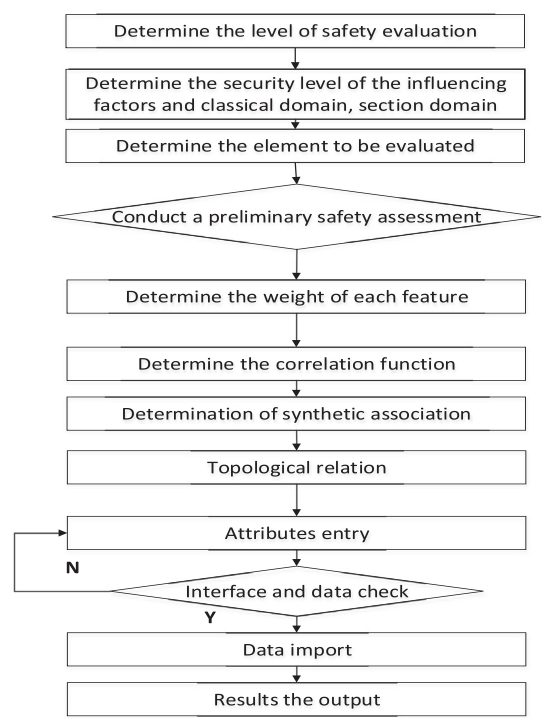

Fig. 4. Safety management evaluation and early warning system based on extension cloud theory.

certainty degree of the value relative to the cloud model. According to the characteristics of comprehensive evaluation of construction safety, each index value to be evaluated is regarded as a cloud drop (Qin et al., 2010), and the calculation of correlation degree is as follows:

Generate a normal random number with mean Ga and standard deviation of Rg, F 'a;

If the index value to be evaluated is m and Pk, then (Mk, Pk) becomes cloud drop;

Use the following formula to calculate the correlation degree:

$$\mathrm{P_k} = \exp[-(\mathrm{m_i} - \mathrm{G_m})^2/(2G'^2_{ak})] \tag{5}$$

where Pk is the index value mi, which belongs to the determination

degree of the cloud, namely the correlation degree between the object index represented by the index value and the object index represented by the cloud, which is called the cloud correlation degree.

In order to ensure the objectivity of the index weight, the weight is determined by the analytic hierarchy entropy weight method. H is assumed to be the Tk index weight obtained by Ahp, and the weight calculated by ahp is modified by using entropy technology, as shown below (Lo et al., 2008):

$$Q_x = E_x C_x / \sum_{x=1}^{n} E_x C_x \tag{6}$$

$$E_x = (1 - I_x) / \sum_{x=1}^{n} (1 - I_x) \tag{7}$$

$$I_x = -\sum_{x=1}^{n} m_{xy} (lnm_{xy}) / \ln a \tag{8}$$

$$m_{xy} = H_{xy} \frac{1}{\sum_{x=1}^{n} H_{xy}} \tag{9}$$

where: Qx is the weight of the x-th index obtained by the analytic hierarchy process entropy weight method; Ex is the information weight of the x-th index; Ix is the entropy value calculated by the x-th index; Hxy is the component of the judgment matrix H by comparing the two indices of the same level. The matrix composed of Qxy forms the comprehensive weight vector U.

2.6. Evaluate the construction safety level

The comprehensive evaluation vector J = UV is obtained by multiplying the comprehensive weight vector U and the comprehensive evaluation matrix V, then the eigenvalue m of the fuzzy rank is:

$$m = J_k L_k \tag{10}$$

where m is the characteristic value of fuzzy rank; Jk is the maximum component value of vector J; Lk is the score of the maximum component corresponding to grade k, and the score of the evaluation grade 15 is 15.

There is a random factor in calculating the cloud correlation degree of m, and multiple operations are required to reduce the randomness, i.e.:

$$G_{mn} = (m_1 + m_2 + ... + m_a)/a \tag{11}$$

$$G_{ma} = \sqrt{\frac{\sum_{k=1}^{a} (m_a - G_{mn}^2)}{a}} \tag{12}$$

where a is the operation number of fuzzy evaluation score, which is taken as 100 times; Mk is the characteristic value of fuzzy rank obtained by the k-th calculation. Gmn is the expected value of multiple operations; Gma is the entropy of multiple operations. The expected value Gma obtained through multiple operations can reflect the overall construction safety condition, and the Gma value can be linked with the grade to judge the safety grade (Teo and Feng, 2010).

The confidence factor O is introduced, and the expression of O is:

$$O = G_{ma} / G_{mn} \tag{13}$$

The evaluation indicators below level 1 all have safety hazards. To accurately find the risk factors, through comprehensive evaluation matrix Z to find the correlation index of cloud, from big to small lined, according to the arrangement of the indexes for risk factors, the number of risk factors (quantity) and numerical equal LioJ construction safety integrated level.

3. Safety management early warning model for construction enterprises

3.1. Model building methods

Based on the principles for the construction of security management early warning mode, the establishment and analysis methods of the safety management early warning model and the safety management evaluation model of construction enterprises are basically the same, except that the provisions of classification, object element to be evaluated, classic domain and section domain are different, many scholars have obtained good research results in this aspect (Chen and Liu, 2016; Chen et al., 2018; Lu et al., 2018; Zhao and Chen, 2015). According to the influencing factors of various accidents, the author divides the safety level of accidents into four grades: no warning, slight warning, intermediate warning and severe warning. The most easy to occur in the construction of the three kinds of accidents is falling, collapse, object, the influence factors of the three kinds of accident is very much also, from the security check can be summarized in the table, according to the main cause of accidents, of each accident can be determined for a review of the matter-element, classical domain and section domain, most only appeared in the accident probability of falling below as an example, its object under evaluation (Hon et al., 2014; Jabori et al., 2013; Xu and Yu, 2014).

The metadata, classical domain and section domain are represented as follows:

Subject matter element:

Classical field:

Joint domain:

The representation of other accident prone elements, classical domain and section domain is similar to the above situation. Warning model compared to the classic section domain and space domain, joint domain of the lower limit value is greater than the classical field similar to the safety management evaluation model of space reasons, according to the safety checklist ratings of the situation and the safety of the construction enterprise management experience, the author adopts the joint domain limit 5% larger than the classical field range.

The calculation process of safety management early warning is similar to that of safety management evaluation. By applying extension theory, the pending evaluation can be analyzed and the categories of vulnerable safety accidents can be determined.

3.2. Correlation analysis

According to the correlation function of the extension set, the correlation degree of each evaluation index on the category grade is:

$$W_{kk(ek)} = \begin{cases} -C(e_k, E_{0bk}) / |E_{0bk}| \\ -C(e_k, E_{0bk}) / (C(e_{k,Eck}) - C(e_k, E_{0bk})) \end{cases} \tag{14}$$

where C () is the distance between the point and the interval.

Considering the weight Wk of each feature Dk, the correlation degree and weight coefficient are synthesized into comprehensive correlation degree:

Table 2
The element to be evaluated.

M	A	Employee safety education	e1
		Three treasures four mouth protection	e2
		The scaffold	e3
		Template engineering	e4
		Equipment	e5

Table 3
The classical domain to be evaluated.

M1	No warning	d1	2	M3	Intermediate warning	d1	1.1
		d2	9.5			d2	6.8
		d3	4.7			d3	3.5
		d4	4.7			d4	3.5
		d5	9.5			d5	7
M2	Minor warning	d1	1.4	M4	Severe warning	d1	0.8
		d2	8			d2	4.5
		d3	4			d3	2.2
		d4	4			d4	2.2
		d5	8.3			d5	4.5

Table 4
The joint domain to be evaluated.

R	(C, D, E)	C	d1	1.2
			d2	6 + −3.5
			d3	3 + −1.7
			d4	3 + −1.7
			d5	6 + −3.5

$$W_b(A) = \sum_{k=1}^{n} W_k U_{bk}(E_k) \tag{15}$$

where B, I and K are all positive integers. A is the subject to be evaluated, Uk is the weight of each feature, and Wkb (Ek) is the correlation degree of evaluation grade of each evaluation object.

Stay out evaluation unit Wb (A) after the correlation of evaluation category if Wb0 (A) = Max {Wb (A)}, so, can determine A B0 classes, and Wb0 numerical size (A) and the relationship between the level can quantitatively reflect the evaluation units, dependent function W (n) value for A review of the matter-element belonging to A certain level of standard scale.

3.3. Case analysis

Taking a construction site as an example, safety evaluation and early warning are carried out according to the evaluation index system. According to the construction safety inspection standard, the safety level of the construction site is divided into four grades: no alarm, light alarm, medium alarm and heavy alarm. Among them, the value range of each index about each grade is the classical domain. The value range of all the warning grades constitutes the section domain. According to the established construction safety evaluation system (Tables 2–4), the evaluation elements for safety early warning can be determined as shown in formula (16):

$$M = (A, D, E) = \begin{bmatrix} \textit{P}\text{eople} & d_1 \\ \textit{M}\text{aterial} & d_2 \\ \textit{E}\text{quipement - }\textit{M}\text{anagement} & d_3 \\ \textit{S}\text{afety - Management} & d_4 \\ \text{Technology} & d_5 \\ \text{Enviroment} & d_6 \end{bmatrix} \tag{16}$$

According to formula (14) and analytic hierarchy process, the weight is: U1 = (0.1, 0.1, 0.16, 0.14, 0.16, 0.1).

The comprehensive correlation can be obtained from formula (15):

W1 = (−0.16, 0.14, −0.26, −0.52)
W2 = (0.25, 0.05, −0.48, −0.64)
W3 = (0.12, 0.05, −0.47, −0.63)
W4 = (−0.16, 0.12, −0.25, −0.5)
W5 = (−0.3, 0.06, −0.05, −0.38)
W6 = (−0.29, 0.18, −0.12, −0.41)

Comprehensive evaluation: W = W0 * U0 = (−0.08, 0.1, −0.3, −0.52)

By the calculated result, relative to the project site 4 warning level correlation respectively 0.08, 0.1, 0.3, 0.52, according to the construction safety early warning model of Wb0 (A) = Max {Wb (A)}, which can determine the project construction site safety early warning level comprehensive evaluation for “light”. Similarly, according to each level indexes on the security warning level comprehensive correlation shows that the project construction site in personnel, materials, equipment, facilities, management, technology and environmental security early warning level 6 “light”, “p”, “p”, “light”, “light” and “light”, explain the project site in doing comparison of materials and equipment safety control in place, but in implementation, technical operation personnel, management and environment aspects are lacking, processing steps should be taken according to the specific circumstances of improvement, in order to avoid the happening of safety accidents.

3.4. Calculation speed analysis

The example analysis shows that the model established in this study has high reliability in prediction accuracy, and the use of distributed computing also greatly increases the calculation speed. In this regard, we compare the computational time of traditional methods and distributed computing under different data volumes to explore the effect of distributed computing on the calculation speed. The advantages of distributed computing are becoming more and more obvious as the amount of data changes from 200 to 800,000 orders of magnitude. The relevant data results are shown in Table 5, Fig. 5 and Fig. 6.

In addition, we compare the computational accuracy of different distributed computing nodes with the manual judgment. The 10 experts independently evaluate the average value and calculate the accuracy of the distributed calculation. The accuracy varies with the calculation is as shown in Fig. 7.

It can be seen from the comparison results that the calculation accuracy of each node is basically kept at a high level except for a large error in calculation accuracy due to calculation delay or other interference factors. This further illustrates the superiority of the distributed computing algorithm designed in this paper.

It can be seen from the analysis in Figs. 8 and 9 that the method adopted in this paper can easily obtain the optimal value of distributed computing, and the results are basically consistent with the real value, which verifies the robustness of the algorithm proposed in this paper and can solve the problems of human factors, fuzziness and randomness in the algorithm.

4. Conclusion

Extend cloud theory has been widely used in computer and engineering fields by virtue of its ability to reason about fuzzy and random factors. This paper has carried out in-depth research and discussion on extension cloud theory and distributed computing methods, and designed a risk assessment suitable for construction industry. The predictive model is an effective exploration of the development of distributed computing.

In view of the present construction safety analysis indicators and the evaluation results problems such as randomness and discreteness and fuzziness, uncertainty reasoning features of the cloud model into matter-element theory of extenics, we build a construction safety comprehensive evaluation model based on extension cloud and

Table 5
Calculation time of different calculation methods under different data quantities.

Data Amount	200	2000	20,000	200,000	800,000
Conventional method	0.62	0.59	5.32	31.36	111.62
Distributed Computing	0.02	0.03	2.09	17.5	97.22

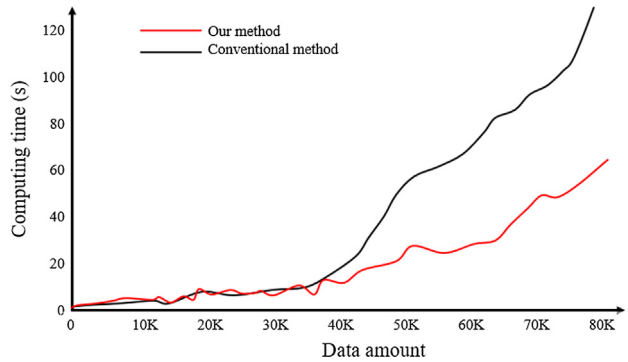

Fig. 5. Calculation time of different calculation methods under different data quantities.

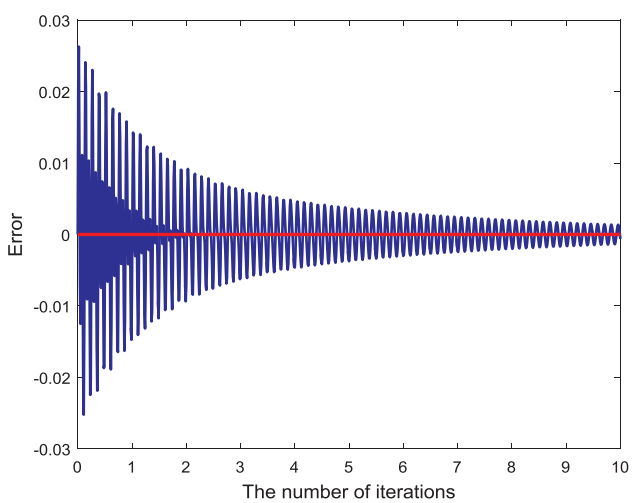

Fig. 6. Calculation error of the method proposed by this paper under different iterations.

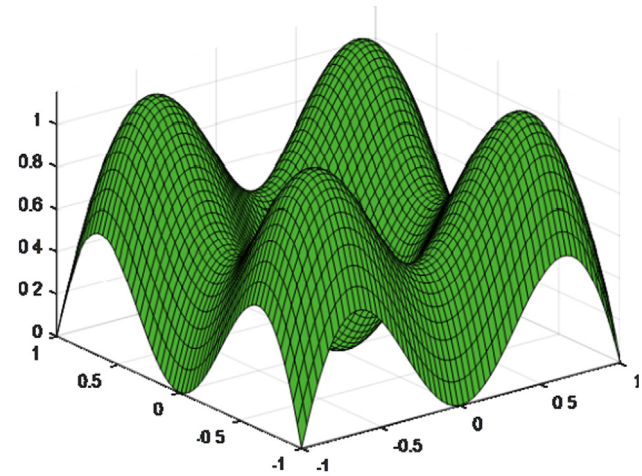

Fig. 8. The optimal value calculated by distributed computing.

distributed computing. Through example, extension cloud model can not only judge the overall state of construction safety and based on the information feedback to identify potential safety problems, grasp the objective existence of the hazards in the construction process, and has an important guiding significance to the improvement of construction safety.

In the future, we will further improve the extension cloud model designed in this paper. We will improve the existing model for the current situation of complex construction environment, many reference factors and artificial experience evaluation in the construction field. In addition, to further enhance the ability of distributed computing, for the construction industry in the construction industry, the environment is poor, equipment conditions are poor, etc., to develop a distributed computing method with low computing performance requirements, to solve the new problems faced in distributed computing.

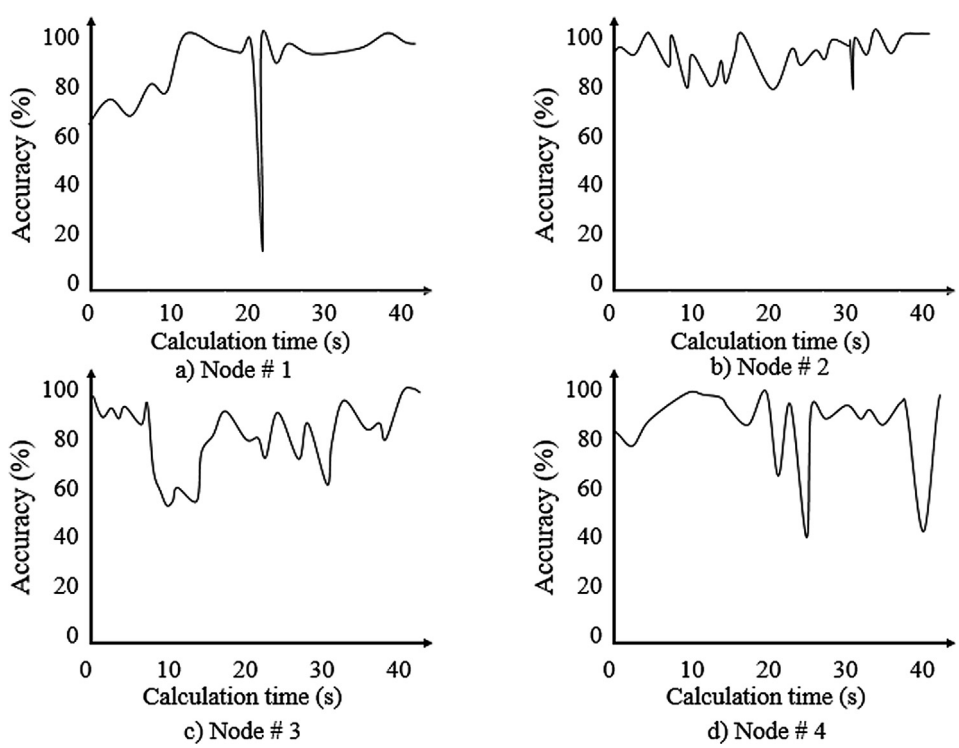

Fig. 7. Comparison between the accuracy of distributed computing nodes and the accuracy of expert evaluation.

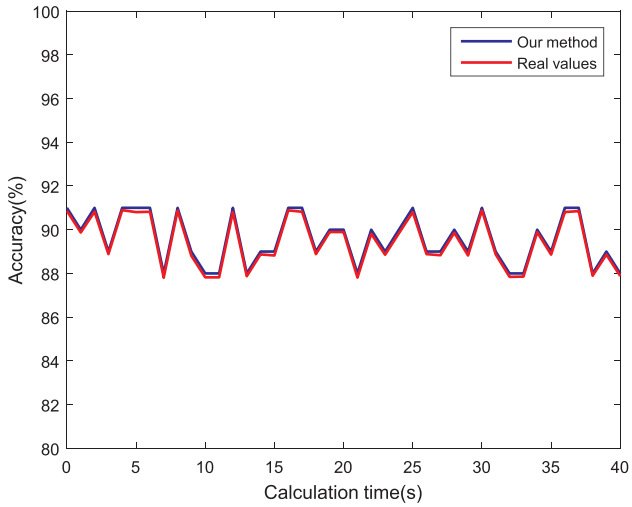

Fig. 9. Comparison between the value obtained by the algorithm in this paper and the real value.

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